ITAI-1371 - Intro to Machine Learning
Reflective Journal

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In my part of Lab 05, I learned how to handle missing numerical data using median imputation. Instead of removing passengers whose ages were missing, I filled those gaps so we could keep their other information in the dataset. I also learned why the median can be a useful choice: unusually high or low ages can affect the mean, while the median is less sensitive to those values. However, the replacement ages are estimates, not the passengers' actual ages. This task helped me understand the importance of checking my results after changing the data. Verifying that no missing ages remained confirmed that the imputation worked and that the column was ready for the next steps in our group's workflow.

Danh Nguyen

What I learned of my part of Lab 5 was answering the three Knowledge Check questions. I learned that we use the median to fill missing ages because extreme values can pull the mean off. I also learned that One-Hot Encoding turns categories into 1s and 0s, since numbering them 0, 1, 2 would make the model think there's an order.

What surprised me is I thought you always had to scale data, but Decision Trees don't need it. They just pick a cutoff like "Age ≤ 30," and scaling doesn't change who falls on each side.

Connection to the full workflow I had to go through everyone's tasks, which showed me how each step builds on the one before it.

Great Inioluwa Olayiwola

Working through this data preparation lab made me realize that preprocessing isn't just a cleanup step, it's where we shape raw information into a format models can actually understand. Raw data is messy and full of missing details. Preparing it properly ensures our machine learning models perform fairly and accurately.

What I Learned & Key Insights

Imputing Missing Data

I used to think deleting rows with missing data was the easiest solution. However, doing that throws away valuable information. When handling missing values in the Age column, using the median made much more sense than the mean. The median isn't pulled off-course by extreme values (like very old or very young passengers), making it a safer pick for skewed data.

Converting Text to Numbers (One-Hot Encoding)

Models operate on numbers, not words. To convert categorical text like Sex or Embarked into numbers, I used One-Hot Encoding. I learned that simply assigning numbers (like 1, 2, 3) creates a fake order or ranking that tricks the model. Creating binary columns (1 or 0) solves this. Using drop_first=True also prevents redundant data by dropping extra columns the model can easily infer.

Scaling Numerical Features

Features like Age (0–80) and Fare (0–512) live on completely different scales. Without scaling, a model might assume higher numbers in Fare are naturally more important. Using StandardScaler puts both features on equal footing centered around a mean of zero. I also realized that distance-based models need scaling, but tree-based models don't because they split data using simple "greater than" comparisons on individual features.

Problem Solving & Technical Lessons

While coding Task 1, I encountered a Pandas FutureWarning because I used inplace=True. I learned that Pandas is deprecating this syntax because chained indexing can create unpredictable copies in memory. I fixed this by using explicit reassignment instead: df['Age'] = df['Age'].fillna(median_age). This taught me the importance of writing future-proof code.

Final Thoughts

My biggest takeaway is that data preparation requires thoughtful decisions at every turn. Choosing how to fill missing values or encode variables directly impacts how well a model learns. Data preparation isn't just administrative work; it forms the core foundation of machine learning.

Karima Sadruddin

What I learned:

My part of the Module 05 Data Preparation lab is Task 3, which focuses on feature scaling. I learned that numerical features can have very different ranges. In the Titanic dataset, Age and Fare are measured on different scales, so one feature can have much larger numbers than the other.

I used StandardScaler from scikit-learn to standardize the Age and Fare columns. Standardization expresses each value in terms of how far it is from that feature's mean, measured in standard deviations. A scaled value of 0 is at the mean, a positive value is above the mean, and a negative value is below the mean.

I learned that scaling is useful for many machine-learning methods that are affected by feature magnitude, but it is not required for every model. For example, Decision Trees generally make splits using thresholds and usually do not need standardized features. This task helped me understand that preprocessing choices should fit the model and the data.

Why verification matters

After scaling, I check that the means of Age and Fare are close to zero and their population standard deviations are close to one. I also check for missing values. These checks help me confirm that the transformation worked as expected instead of assuming that the code is correct just because it ran.

Connection to the full workflow

Feature scaling comes after earlier preparation steps, such as handling missing Age values and encoding categorical columns. The steps must be completed in order because Task 3 depends on the prepared dataset from Task 2.

Overall, I learned that careful preprocessing and checking results are important parts of building a reliable machine-learning workflow.

Quan Thach

Going into this lab, I thought data preparation would be the boring part of machine learning, something to rush through before the real modeling begins. The Titanic notebook changed that opinion quickly. When I checked for missing values, I found 177 missing ages, 687 missing cabin entries and two missing embarkation ports. Seeing those numbers made it clear that a model fed this raw table would either crash or quietly learn from bad information.

Task 1, imputing the Age column, was the most intuitive. The median is safer than the mean because a few very old passengers can pull the average upward, and filling the gaps kept all 891 rows instead of throwing hundreds away. The starter code used fillna with inplace=True, which can behave unpredictably in newer versions of pandas, so I assigned the result back to the column instead. That left zero missing values in Age.

Task 2, one-hot encoding, taught me the most. Models need numbers, but labeling categories as 1, 2 and 3 would make the model read a ranking into groups that have no natural order. Splitting each category into its own yes-or-no column avoids that. The drop_first=True option confused me at first, until I realized that if Sex_male is False, the passenger must be female, so keeping both columns repeats the same information.

Task 3, feature scaling, showed me how different the columns really are. Age runs to about 80 while Fare passes 500, so an unscaled model could treat Fare as more important just because its numbers are bigger. I scaled only Age and Fare, not the encoded columns or the Survived target.

The knowledge check tied it together, especially the decision tree question. Trees split on thresholds, so scaling isn't needed for them, which showed me that preprocessing depends on the model rather than following a fixed checklist. Overall, I walked away more patient with the cleaning stage and more confident explaining why each step is done, not just how to code it.

This version is about 310 words, which is under the 500-word minimum you set earlier. If you still need to meet that requirement, I can trim it to something in between.